ECE 340: Semiconductor Devices

Lecture 19: P-N Junction

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Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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(Recap) Band Diagrams With Electric Fields

- As the electron is accelerated in the field, it loses potential energy and gains kinetic energy
- When an electric field is present, the bands are tilted to reflect the loss in potential energy as the electron moves in the direction opposite to the field
- The electric field can be externally applied or internally generated by doping gradients

$\mathcal{V}(x) \equiv$ Electrostatic Potential

$E(x) \equiv$ Electron Potential Energy

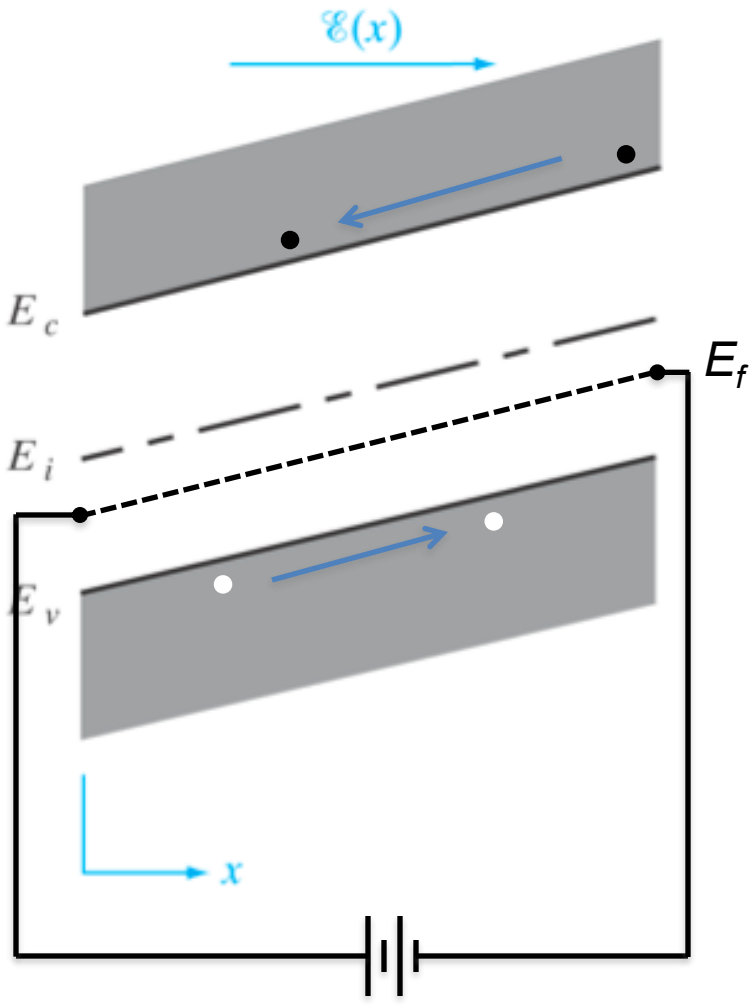
$\mathcal{E}(x) \equiv$ Electric Field

$$\mathcal{V}(x) = \frac{E(x)}{(-q)} \text{ and } \mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx}$$

Choosing E_i as a spatial reference:

$$\mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx} = -\frac{d}{dx}\left[\frac{E_i}{(-q)}\right] = \frac{1}{q}\frac{dE_i}{dx}$$

Externally Applied Field:



(Recap) Band diagram of graded doping

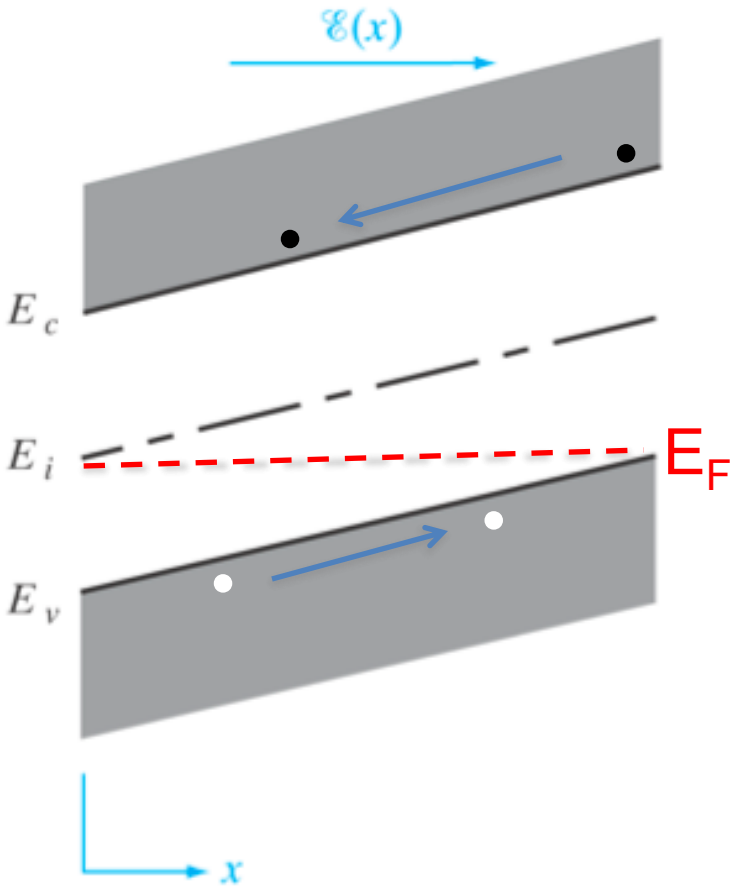
- E field can be generated internally.
- How can we generate the linear field?

$$E_i(x) = E_i(0) + q\mathcal{E}_0 x, \qquad \mathcal{E}_0 = \frac{1}{q}\frac{dE_i}{dx} > 0.$$

$$p_0(x) = n_i e^{|E_i(x) - E_F|/k_B T}$$

- Exponentially increasing doping concentration along *x* can generate linear field.
- Fermi level is invariant (equilibrium) → there is no net current (why?)

Internally Generated Field:



We can also have tilted bands with no externally applied potential!

(Recap) Derivation of Einstein Relationship

$$J_p(x) = qp(x)\mu_p\mathcal{E}(x) - qD_p\frac{dp(x)}{dx} = 0$$

$$\text{so } \mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{p(x)}\frac{dp(x)}{dx}$$

$$p_0(x) = n_i e^{(E_i(x)-E_F(x))/kT}.$$

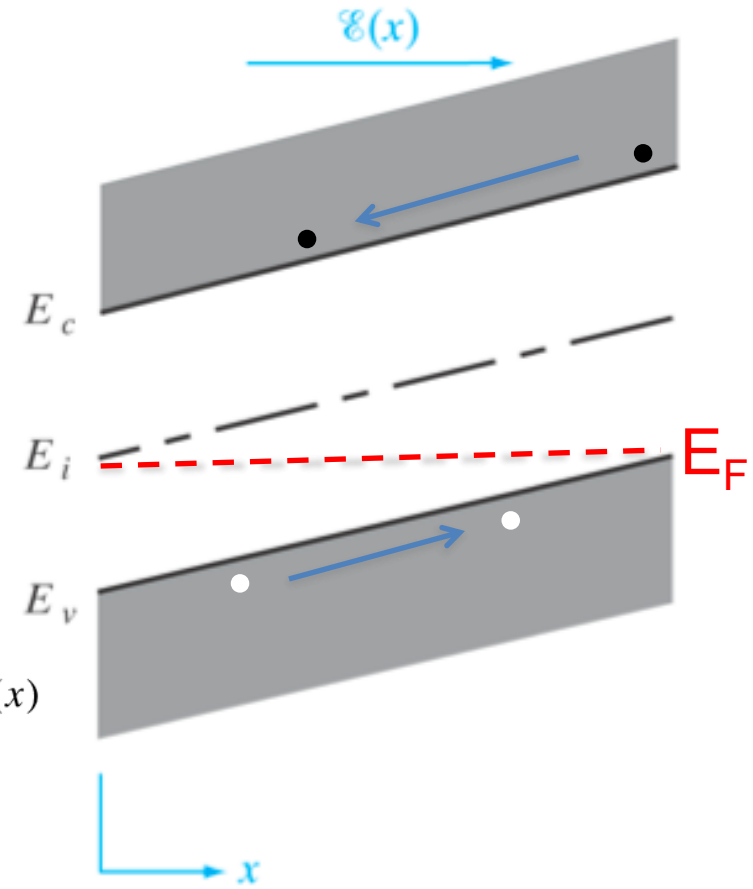
$$\text{using } \frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x)$$

$$\frac{dp_0}{dx} = n_i e^{(E_i-E_F)/kT}\frac{d}{dx}\left[\frac{E_i - E_F}{kT}\right]$$

$$\mathcal{E}(x) = \frac{D_p}{\mu_p}\frac{1}{kT}\left(\frac{dE_i}{dx} - \frac{dE_F}{dx}\right) = \frac{D_p}{\mu_p}\frac{1}{kT}(q\mathcal{E}(x) - 0) \text{ since } \frac{1}{q}\frac{dE_i}{dx} = \mathcal{E}(x)$$

$$\text{so } 1 = \frac{D_p}{\mu_p}\frac{q}{kT} \text{ and } \boxed{\frac{D_p}{\mu_p} = \frac{kT}{q}}$$

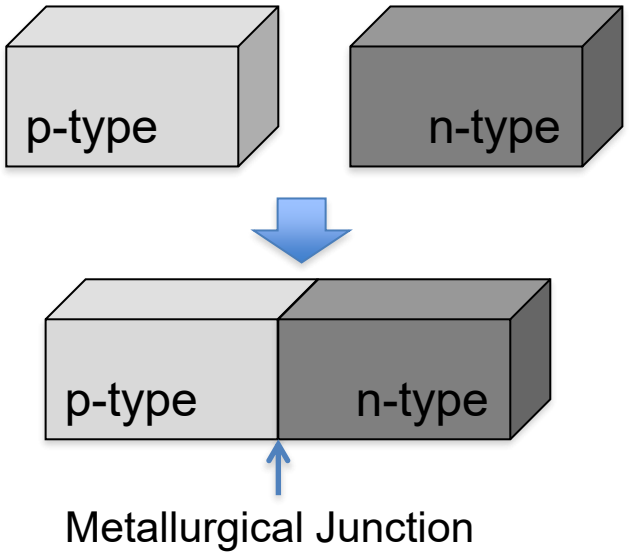
Internally Generated Field:



We can also have tilted bands with no externally applied potential!

Definitions

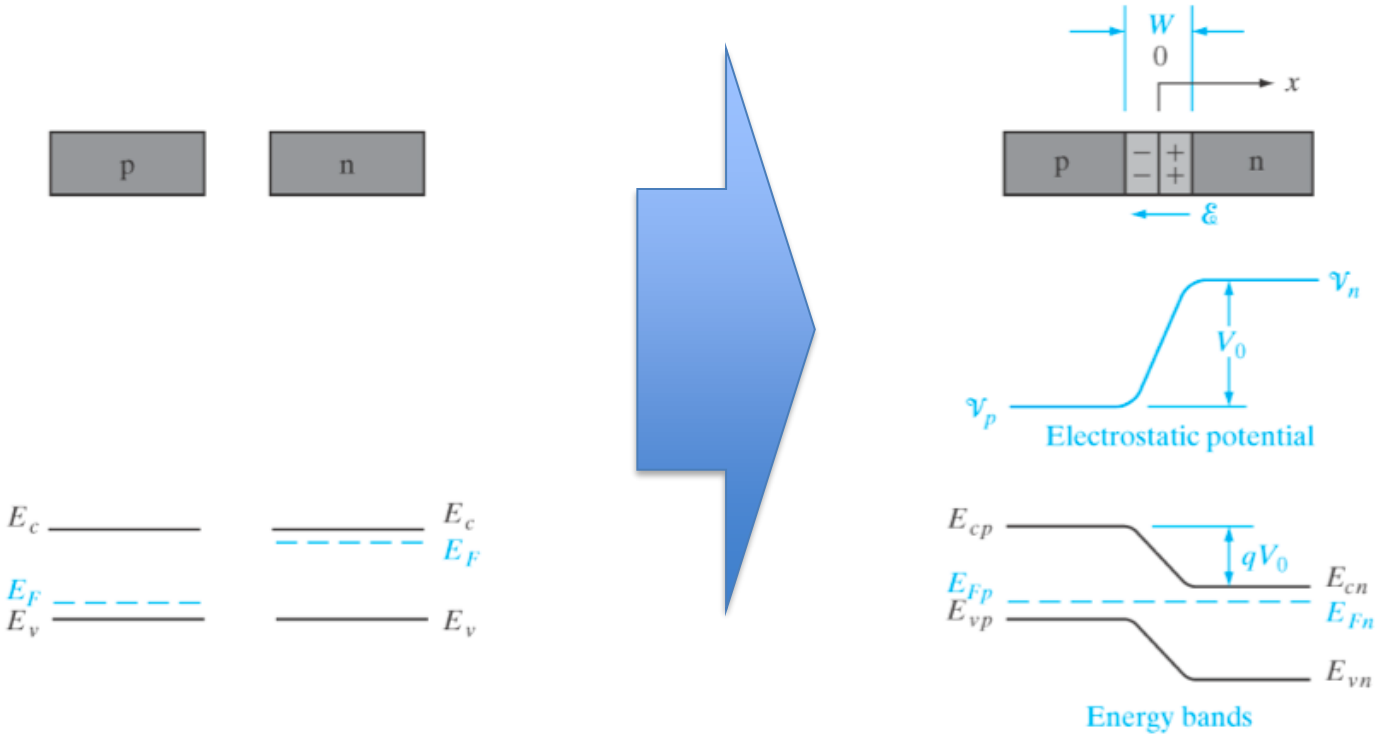
- ***Metallurgical Junction:*** The specific point at which the transition between p-type and n-type material occurs



- ***Step Junction:*** A junction in which there is an **abrupt transition** in the material properties (doping concentration & type) between each side, with each side uniform
- ***Graded Junction:*** A junction in which the material properties (compensated doping concentration & type) change in the region of the junction

Equilibrium Condition

Equilibrium pn Junction



Equilibrium pn Junction

- In equilibrium, there is no net current flow across the junction, so:
  - J_p(Drift)+J_p(Diffusion)=0
  - J_n(Drift)+J_n(Diffusion)=0
- The electric field balances the diffusion current
- The potential difference $V_o=V_n-V_p$ develops in the direction opposite to the electric field E
- V_o is the "***contact potential***" which is a ***built-in potential*** barrier
  - Contact potential cannot be measured
- W is the "***transition region***"
  - Also called "depletion region" and "space charge region"
- We can calculate V_o from the separation in the Fermi levels (this will be shown later)

